ER DIAGRAM

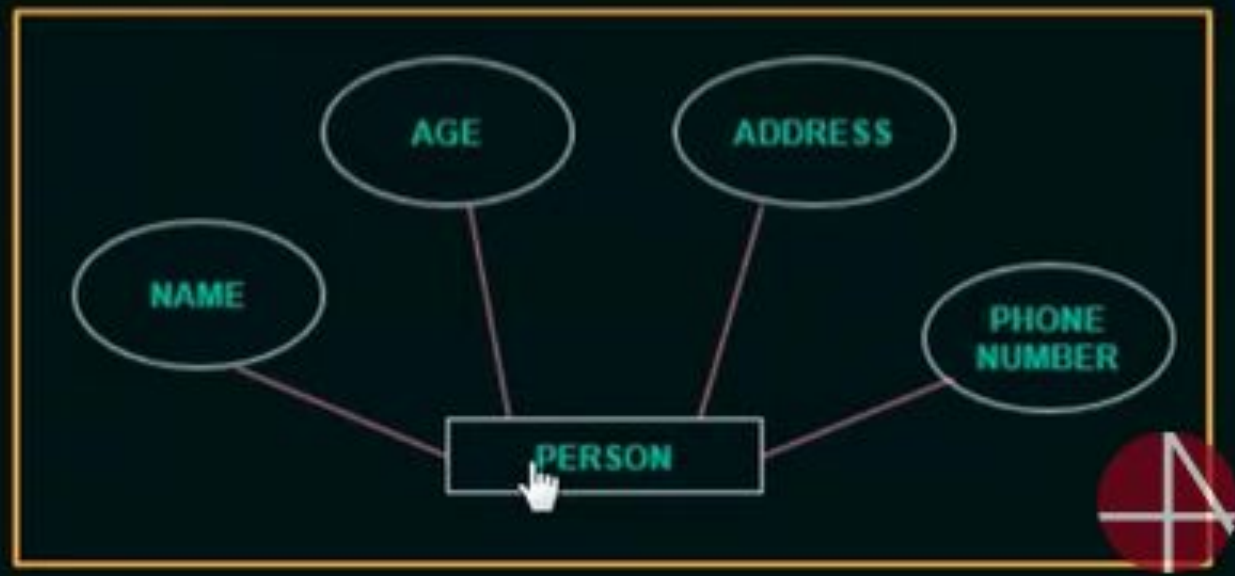
❖ Terminologies

★ Entity:

- A "thing" in the real world with an independent existence.
- May be an object with physical existence (ex: house, person) or with a conceptual existence(ex: course, job).

★ Attributes:

- Properties that describe the entities.



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❖ Terminologies

Composite Attributes

★ Can be divided into further parts.

★ Ex: Name → First Name, Middle Name, Last Name

Address

Simple Attributes

★ Cannot be divided further.

★ Ex: Weight → cannot be further divided.

Age, Salary



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❖ Terminologies

Single-Valued Attributes

- ★ Have a single value for a particular entity.
- ★ Ex: Age → single-valued attribute of a person.

Height, Weight, Salary

Multivalued Attributes

- ★ Can have set of values for a particular entity.
- ★ Ex: College degree, languages known → multivalued attributes of a person.



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❖ Terminologies

Derived Attributes

★ Can be derived from other attributes.

★ Ex: Age → can be derived from date of birth.

Stored Attributes

★ From which the value of other attributes are derived.

★ Ex: BirthDate of a person



❖ Terminologies

★ Complex Attributes:

- Has multivalued & composite components in it.
- Multivalued attributes → represented within '{ }'.
- Composite attributes → represented within '( )'.
- Ex: {CollegeDegrees(College, Year, Degree, Field)}

★ Null Values:

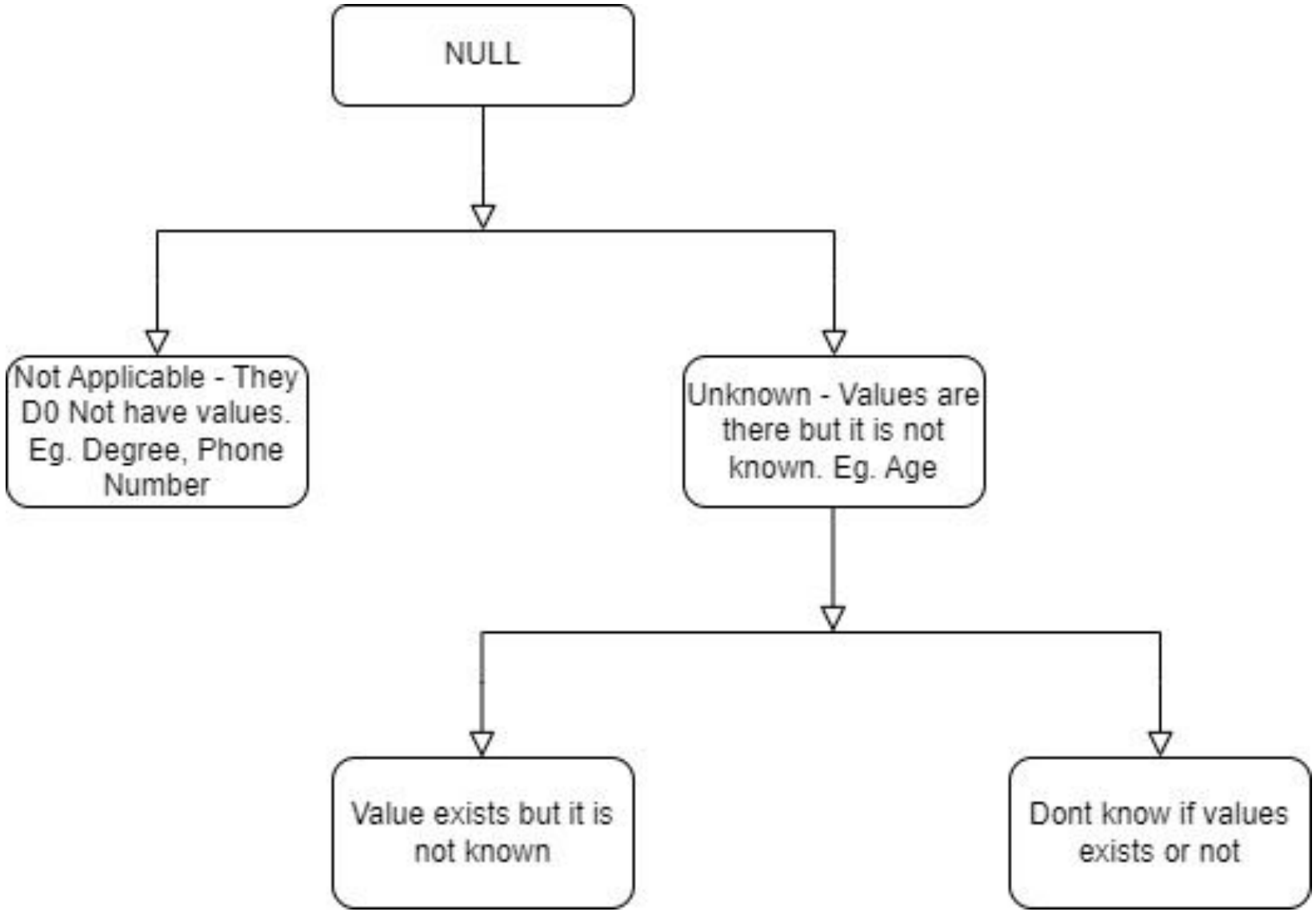
- Null is something which is not applicable or unknown.

They do not have values.
Eg. Degree, Phone number

Values are there but it is not known . Eg Age

STUDENT_ID	NAME	AGE	PHONE_NO
1	HARRY	20	123546897
2	JEREMY	22	NULL
3	JOSHUA	NULL	2525468912

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NULL
Not Applicable - They D0 Not have values. Eg. Degree, Phone Number
Unknown - Values are there but it is not known. Eg. Age
Value exists but it is not known
Dont know if values exists or not

❖ Terminologies

★ Entity Type:

- A collection of entities that have the same attributes.
- Ex: STUDENT

STUDENT

STUDENT_ID	NAME	AGE
1	HARRY	20
2	JEREMY	22
3	JOSHUA	18

★ Entity Set:

- Collection of entities of a particular entity type at a point in time.





❖ Terminologies

★ Key Attribute:

- That attribute that is capable of identifying each entity uniquely.
- Ex: Roll number of a student

STUDENT

STUDENT_ID	NAME	AGE
1	PRIYANKA	20
2	JEREMY	21
3	PRIYANKA	20

★ Value Set of Attributes:

- The set of values that can be assigned to an attribute.

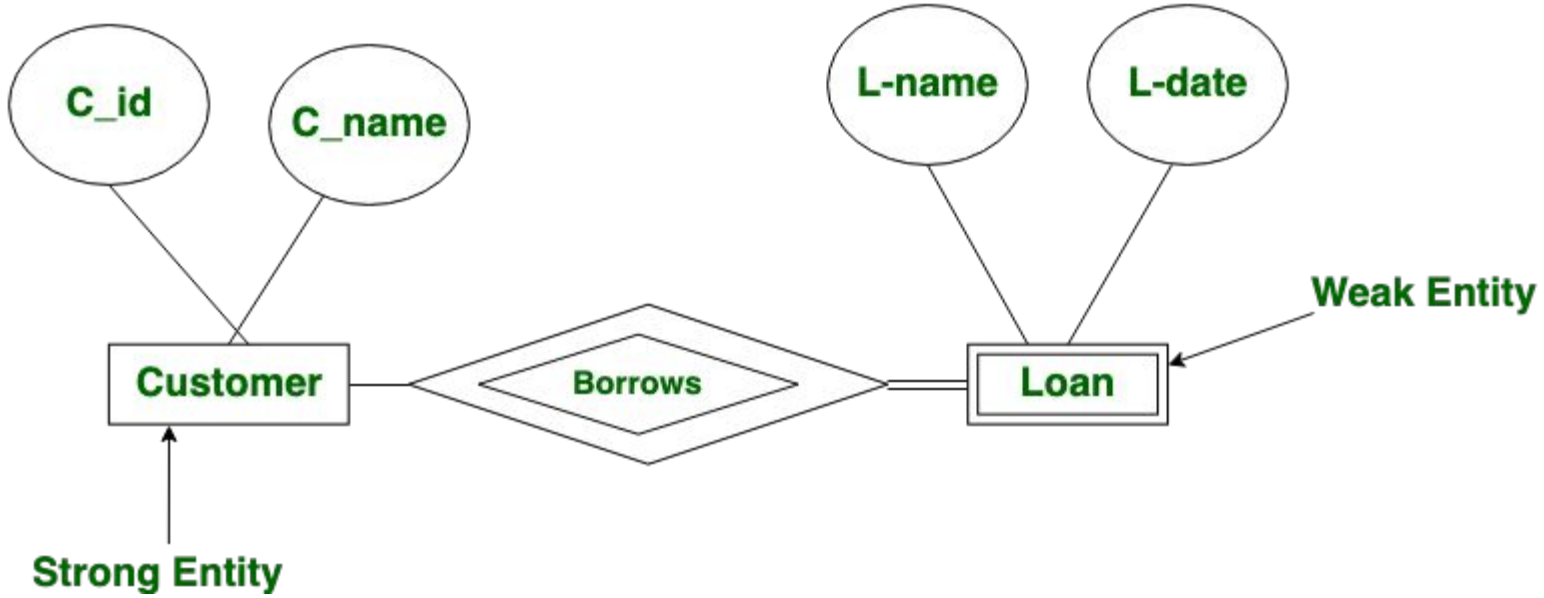


❖ Weak Entity Types

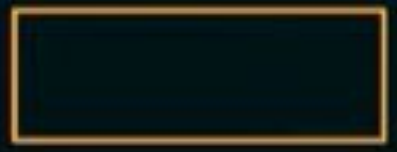
- Entity types that do not have key attributes of their own.
- Identified by relating to another entity type called the identifying or the owner entity type.
- Relationship between weak entity type to its owner → identifying relationship.

Strong Entity:
A strong entity is not dependent on any other entity in the schema. A strong entity will always have a primary key. Strong entities are represented by a single rectangle. The relationship of two strong entities is represented by a single diamond.
Various strong entities, when combined together, create a strong entity set.

Weak Entity:
A weak entity is dependent on a strong entity to ensure its existence. Unlike a strong entity, a weak entity does not have any primary key. It instead has a partial discriminator key. A weak entity is represented by a double rectangle. The **relation between one strong and one weak entity is represented by a double diamond.** This relationship is also known as an **identifying relationship.**



Symbols used in ER Diagram:

- Entity → 
- Weak Entity → 
- Attribute → 

Foreign Key - Use dotted underline, Use the abbreviation FK inside parenthesis

- Key Attribute → 

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❖ Symbols used in ER Diagram:

★ Multivalued Attribute



★ Composite Attribute



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❖ Symbols used in ER Diagram:

★ Derived Attribute





Strong Relationship

A strong relationship shows a direct link between two entities.

★ Identifying Relationship



Relationship between Weak entity type and its owner

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❖ Sample Database Application

★ Let us see an example database application, called COMPANY.

★ Requirements gathered:

- Company is organized into departments. Each department has a unique name, unique number & a particular employee who manages the department. We also keep track of the start date of the manager. A department may have several locations.

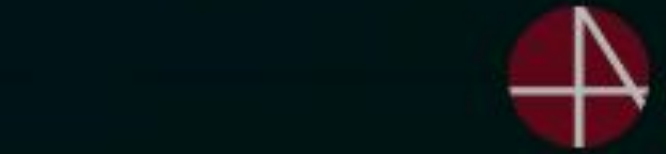


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❖ Sample Database Application

★ Requirements gathered:

- A department controls number of projects, each of which has a unique name, unique number and a single location.
- Employee details → name, SSN, sex, salary. We keep track of number of hours per week on each project.
- Keep track of each employee's dependents (first name, sex, relationship to the employee).



❖ Initial Conceptual Design of COMPANY Database

★ We can identify 4 entity types based on the requirements:

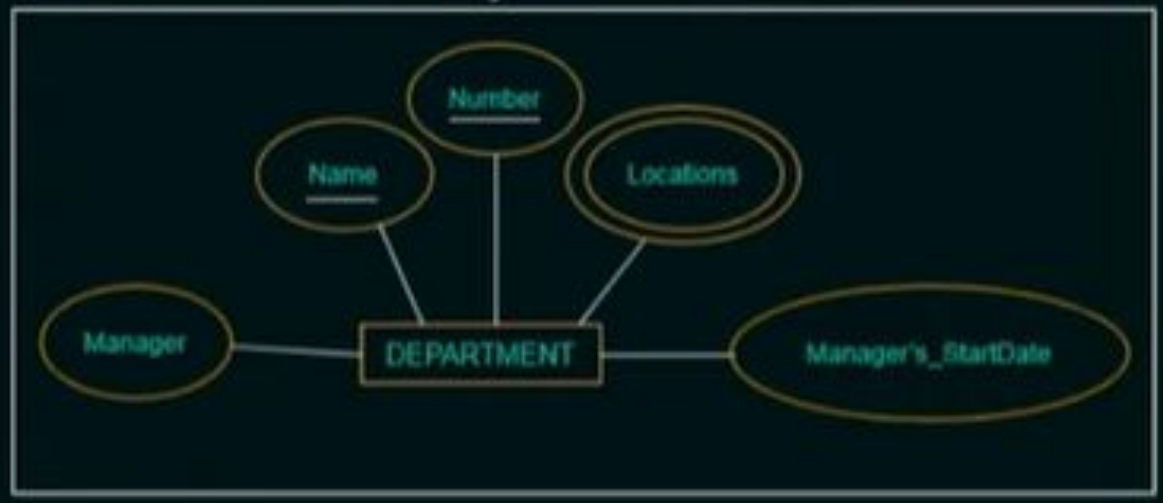
❖ Initial Conceptual Design of COMPANY Database

★ We can identify 4 entity types based on the requirements:

- DEPARTMENT

  Name, Number, {Locations},

  Manager, ManagerStartDate



❖ Initial Conceptual Design of COMPANY Database

★ We can identify 4 entity types based on the requirements:

- DEPARTMENT

  Name, Number, {Locations},

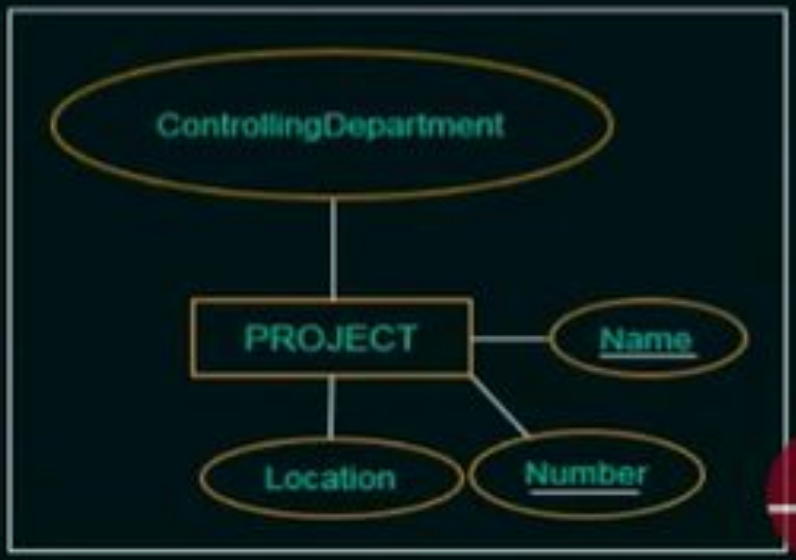
  Manager, ManagerStartDate



- PROJECT

  Name, Number, Location,

  ControllingDepartment



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❖ Initial Conceptual Design of COMPANY Database

★ We can identify 4 entity types based on the requirements:

- EMPLOYEE

  Name(FName, MName, LName),

  SSN, Sex, Salary, BirthDate,

  Department, {WorksOn (Project,

  Hours))



❖ Initial Conceptual Design of COMPANY Database

★ We can identify 4 entity types based on the requirements:

- EMPLOYEE

  Name(FName, MName, LName),

  SSN, Sex, Salary, BirthDate,

  Department, {WorksOn (Project,

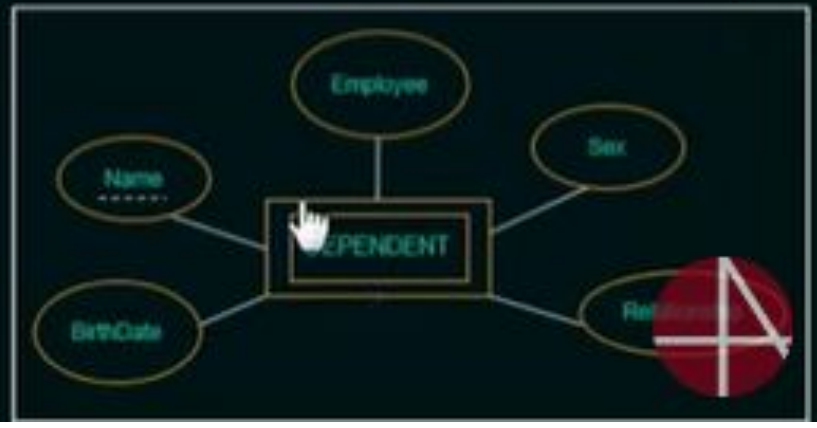
  Hours))



- DEPENDENT

  Employee, DependentName, Sex,

  BirthDate, Relationship



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1. Library Management System

•Entities:

- **Book**: Book_ID (PK), Title, ISBN, Author_ID (FK), Publisher_ID (FK)
- **Author**: Author_ID (PK), Name, Country
- **Publisher**: Publisher_ID (PK), Name, Address
- **Member**: Member_ID (PK), Name, Membership_Date
- **Loan**: Loan_ID (PK), Book_ID (FK), Member_ID (FK), Loan_Date, Return_Date

2. E-commerce System

•Entities:

- **Product**: Product_ID (PK), Name, Price, Category_ID (FK)
- **Category**: Category_ID (PK), Category_Name
- **Customer**: Customer_ID (PK), Name, Email, Address
- **Order**: Order_ID (PK), Order_Date, Customer_ID (FK)
- **Order_Item**: Order_Item_ID (PK), Order_ID (FK), Product_ID (FK), Quantity

3. University Database

- **Entities:**
  - **Student**: Student_ID (PK), Name, Major, Enrollment_Date
  - **Course**: Course_ID (PK), Course_Name, Credits
  - **Instructor**: Instructor_ID (PK), Name, Department
  - **Department**: Department_ID (PK), Department_Name
  - **Enrollment**: Enrollment_ID (PK), Student_ID (FK), Course_ID (FK), Enrollment_Date

4. Online Food Delivery System

- **Entities:**
  - **Restaurant**: Restaurant_ID (PK), Name, Address
  - **Customer**: Customer_ID (PK), Name, Phone, Address
  - **Order**: Order_ID (PK), Order_Date, Customer_ID (FK)
  - **Food_Item**: Food_Item_ID (PK), Name, Price, Restaurant_ID (FK)
  - **Order_Detail**: Order_Detail_ID (PK), Order_ID (FK), Food_Item_ID (FK), Quantity

❖ Relationships

- Relationship → association among 2 or more entities.
- Ex: teacher teaches student

  ↓

  relationship

★ Degree of Relationship: Denotes the number of entity types that participate in a relationship.

1. Unary relationship:
   - ➢ Exists when there is an association with only one entity.



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❖ Relationships

★ Degree of Relationship:

2. Binary relationship:
   - ➢ Exists when there is an association among two entities.



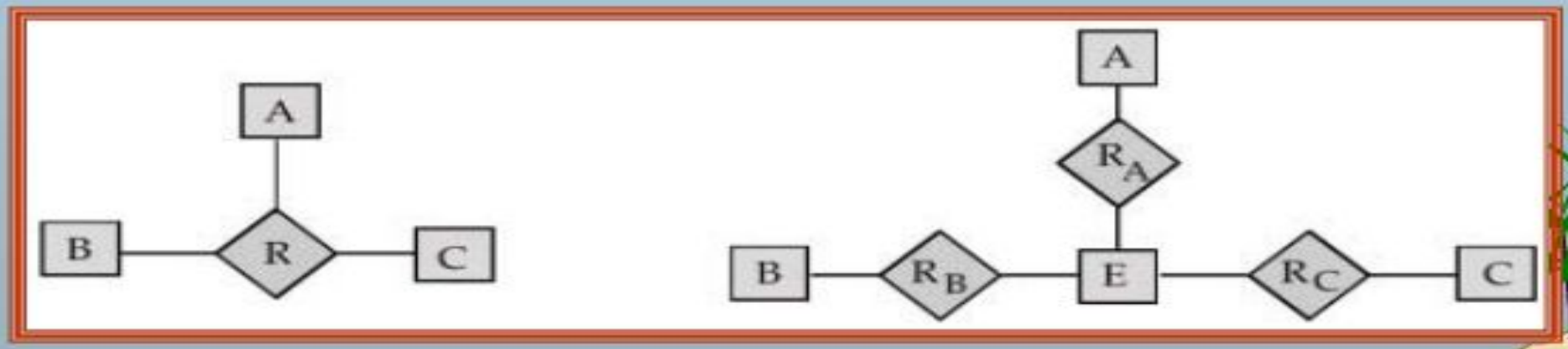
3. Ternary relationship:
   - ➢ Exists when there is an association among three entities.



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Converting Non-Binary Relationships to Binary Form

- In general, any non-binary relationship can be represented using binary relationships by creating an artificial entity set.
  - Replace R between entity sets A, B and C by an entity set E, and three relationship sets:
    1. R_A, relating E and A
    2. R_B, relating E and B
    3. R_C, relating E and C
  - Create a special identifying attribute for E
  - Add any attributes of R to E
  - For each relationship (a_i , b_i , c_i) in R, create
    1. a new entity e_i in the entity set E
    2. add (e_i , a_i) to R_A
    3. add (e_i , b_i) to R_B
    4. add (e_i , c_i) to R_C



❖ Relationships

★ Relationship Constraints:

1. Cardinality Ratio

➢ Maximum number of relationship instances that an entity can participate in.

➢ Possible cardinality ratios for binary relationship → 1:1, 1:N, N:1, M:N.



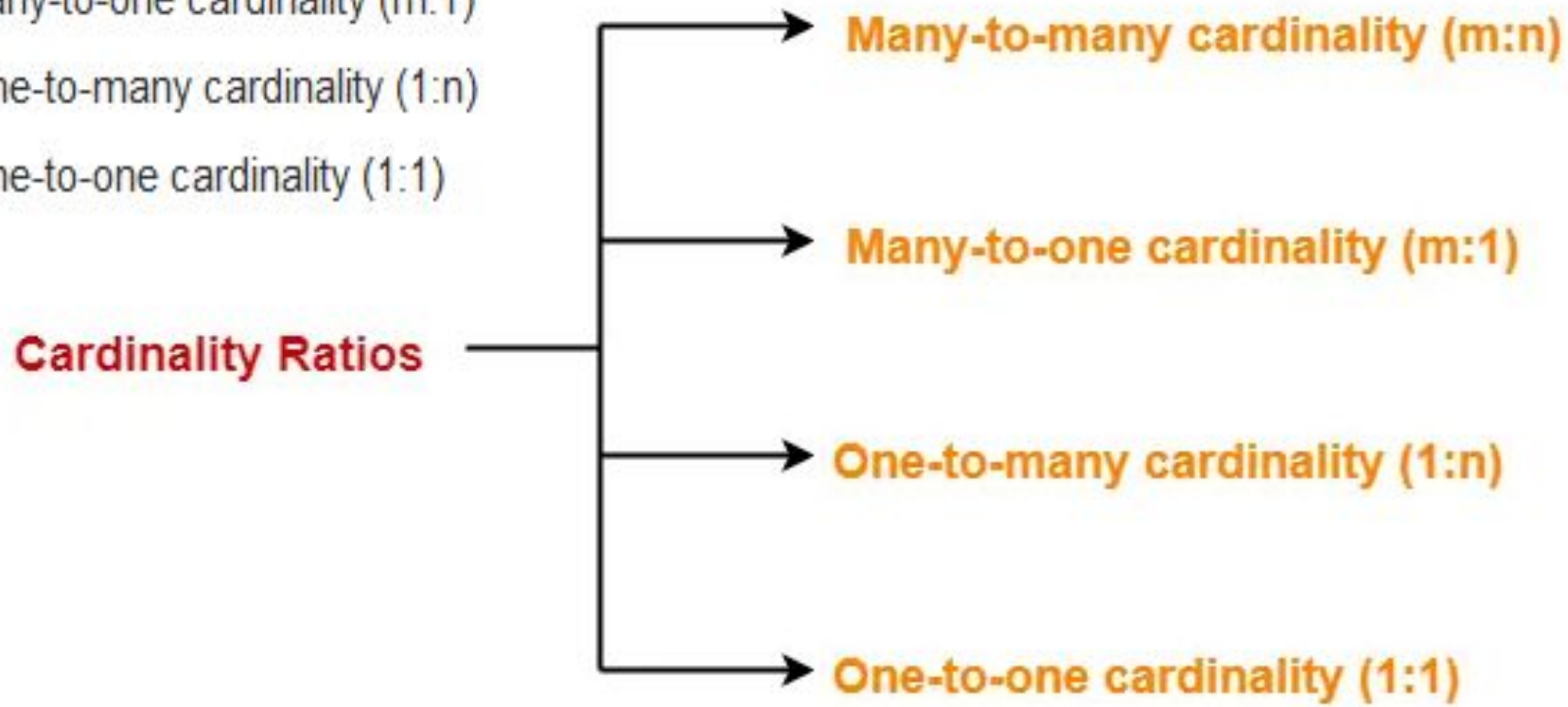
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6. For Cardinality Constraints / Ratios-

Cardinality constraint defines the maximum number of relationship instances in which an entity can participate.

There are 4 types of cardinality ratios-

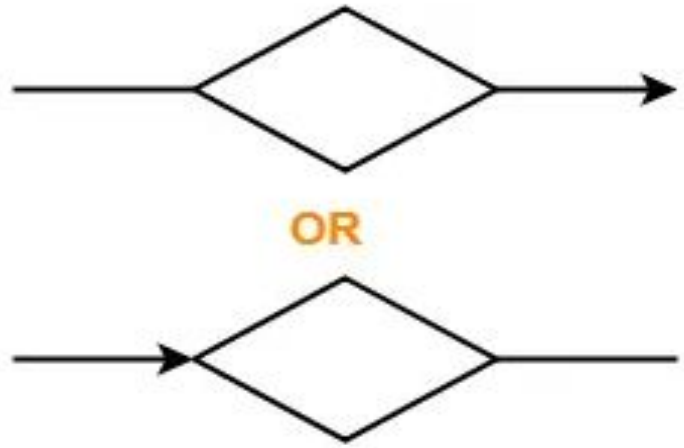
1. Many-to-many cardinality (m:n)
2. Many-to-one cardinality (m:1)
3. One-to-many cardinality (1:n)
4. One-to-one cardinality (1:1)





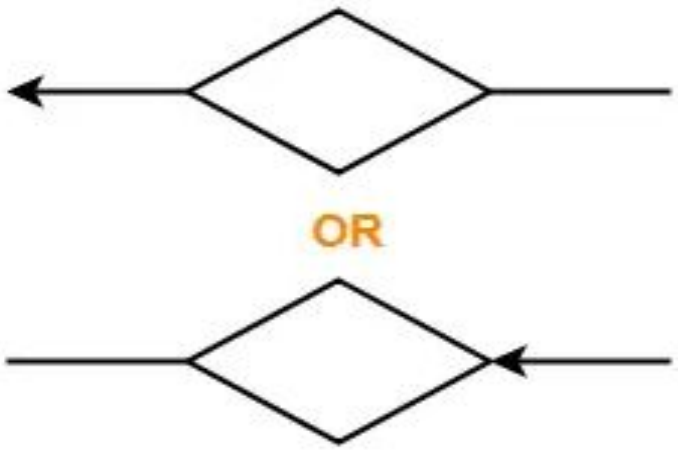
Many-to-Many relationship

(m:n)



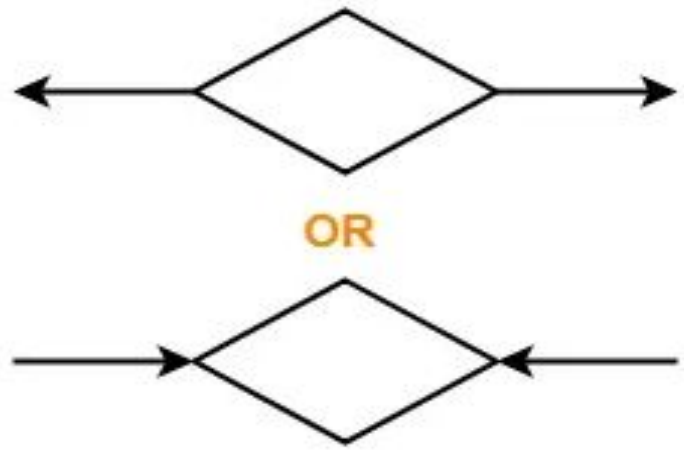
Many-to-One relationship

(m:1)



One-to-Many relationship

(1:n)



One-to-One relationship

(1:1)

1. Many-to-Many Cardinality-

By this cardinality constraint,

- An entity in set A can be associated with any number (zero or more) of entities in set B.
- An entity in set B can be associated with any number (zero or more) of entities in set A.

Symbol Used-



Cardinality Ratio = m : n

Example-

Consider the following ER diagram-



Many to Many Relationship

Here,

- One student can enroll in any number (zero or more) of courses.
- One course can be enrolled by any number (zero or more) of students.

2. Many-to-One Cardinality-

By this cardinality constraint,

- An entity in set A can be associated with at most one entity in set B.
- An entity in set B can be associated with any number (zero or more) of entities in set A.

Symbol Used-



OR



Cardinality Ratio = m : 1

Example-

Consider the following ER diagram-



Many to One Relationship

Here,

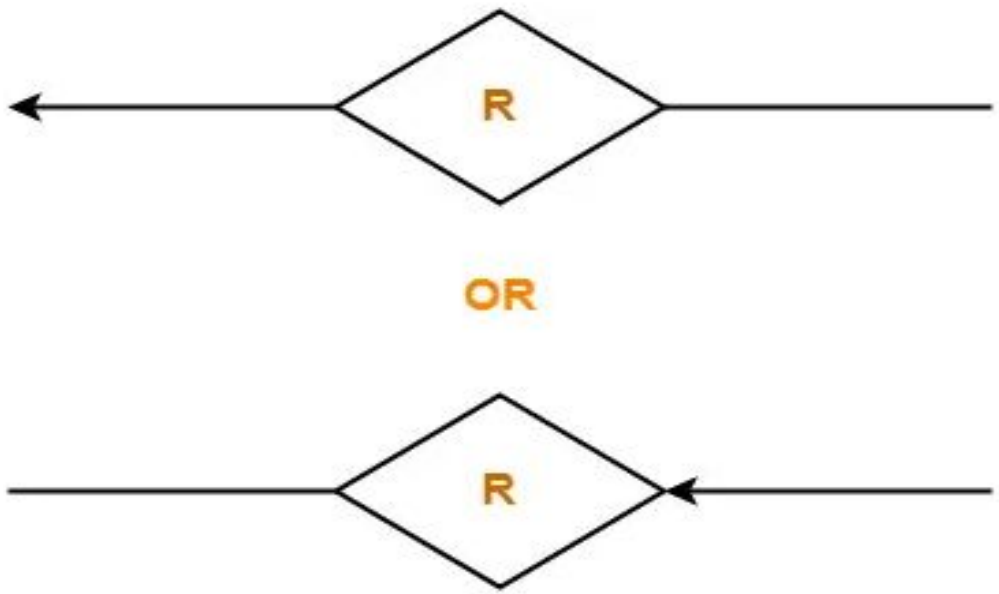
- One student can enroll in at most one course.
- One course can be enrolled by any number (zero or more) of students.

3. One-to-Many Cardinality-

By this cardinality constraint,

- An entity in set A can be associated with any number (zero or more) of entities in set B.
- An entity in set B can be associated with at most one entity in set A.

Symbol Used-



Cardinality Ratio = 1 : n

Example-

Consider the following ER diagram-



One to Many Relationship

Here,

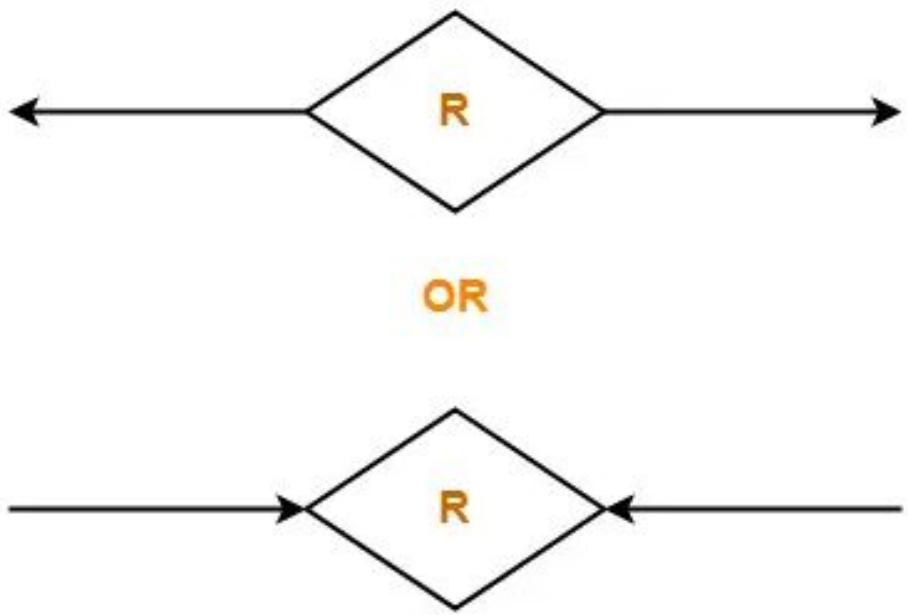
- One student can enroll in any number (zero or more) of courses.
- One course can be enrolled by at most one student.

4. One-to-One Cardinality-

By this cardinality constraint,

- An entity in set A can be associated with at most one entity in set B.
- An entity in set B can be associated with at most one entity in set A.

Symbol Used-



Cardinality Ratio = 1 : 1

Example-

Consider the following ER diagram-



One to One Relationship

Here,

- One student can enroll in at most one course.
- One course can be enrolled by at most one student.

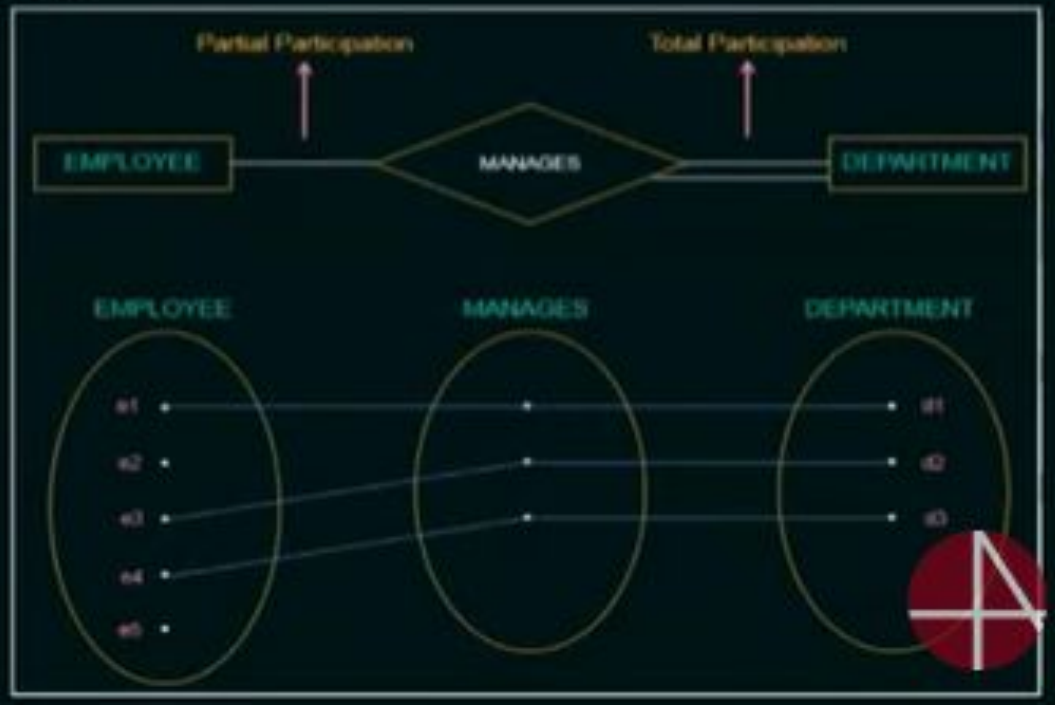
❖ Relationships

★ Relationship Constraints:

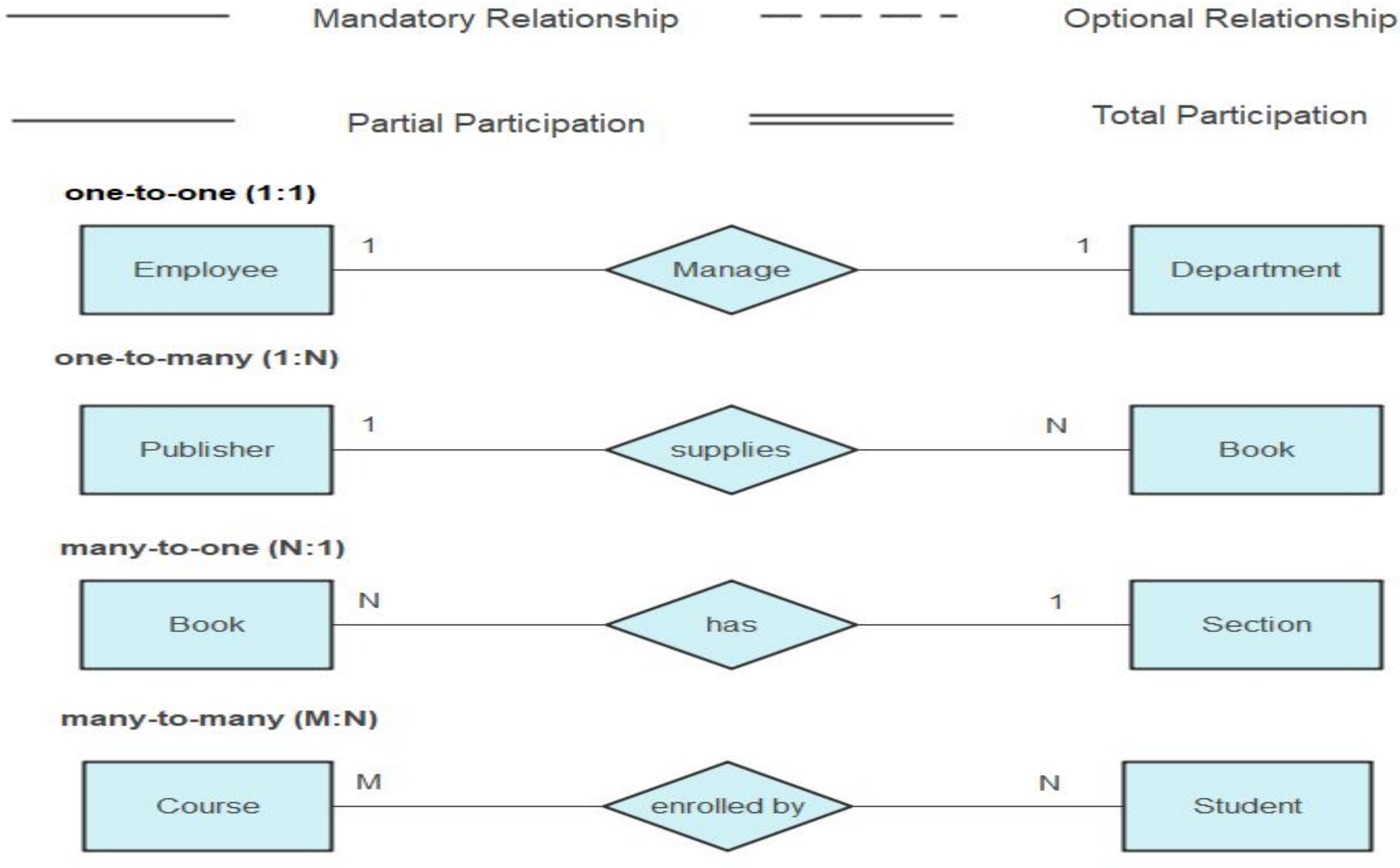
2. Participation Constraints

➢ Specifies whether existence of an entity depends on its being related to another entity.

➢ 2 types: Total participation & Partial participation.



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Mandatory Relationship
Optional Relationship
Partial Participation
Total Participation
one-to-one (1:1)
Employee
1
Manage
1
Department
one-to-many (1:N)
Publisher
1
supplies
N
Book
many-to-one (N:1)
Book
N
has
1
Section
many-to-many (M:N)
Course
M
enrolled by
N
Student

❖ Relationships

★ Attributes of Relationship Types

- Attributes of 1:1 or 1:N relationship types can be migrated to one of the participating entity types.

  → In 1:1 relationship type, attributes can be migrated to either of the entity types.

  → In 1:N or N:1 relationship type, attributes are migrated only to the entity type on the N-side of the relationship.

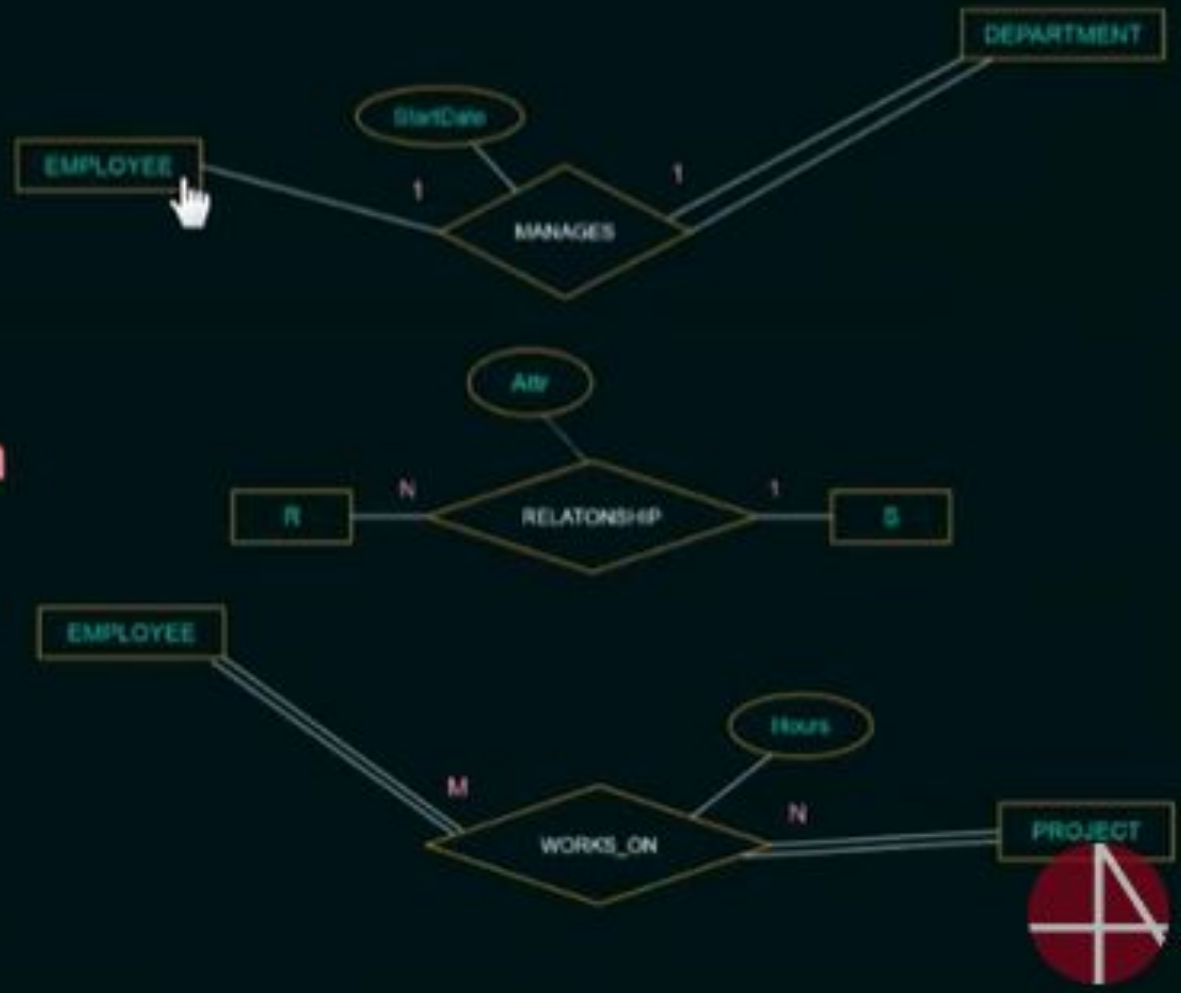


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❖ Relationships

★ Attributes of Relationship Types

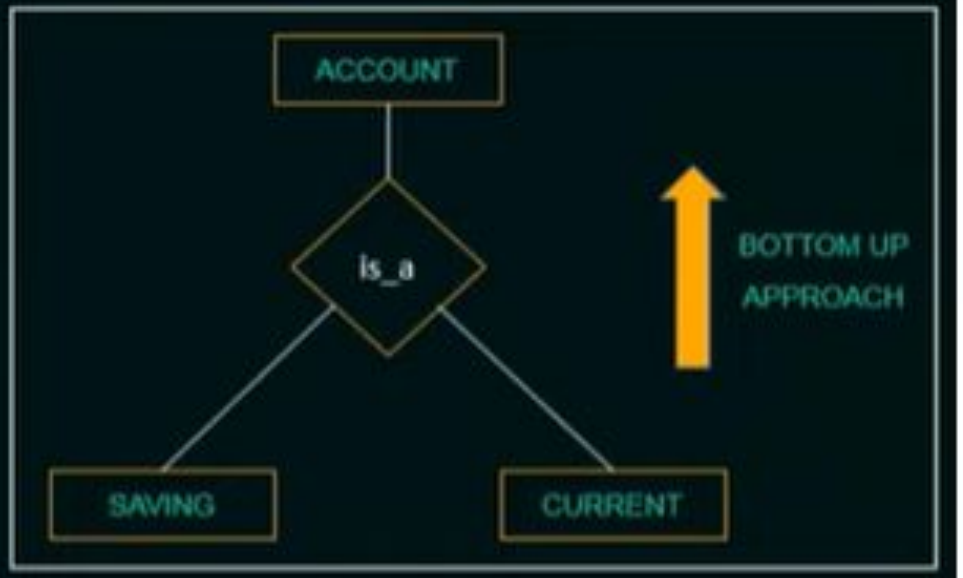
- In M:N relationship type, some attributes can be determined by a combination of participating entities.



❖ Enhanced ER Model

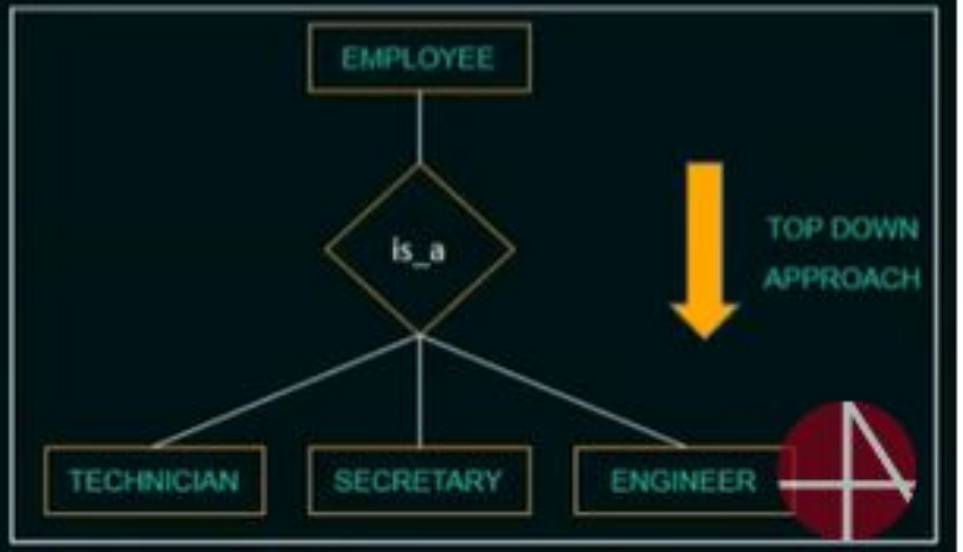
★ Generalization:

- **Bottom-up approach** where two lower level entities combine to form a higher level entity.



★ Specialization:

- **Top-down approach** where it defines set of subclasses of an entity type.



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ER Diagram for COMPANY Database



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Reference Link

https://www.edrawsoft.com/er-diagram-symbols.html

https://www.gatevidyalay.com/er-diagrams/

Assignment – Draw an ER Diagram for University Database

https://www.tutorialspoint.com/draw-an-er-diagram-for-a-university-database-application-dbms

Exercises

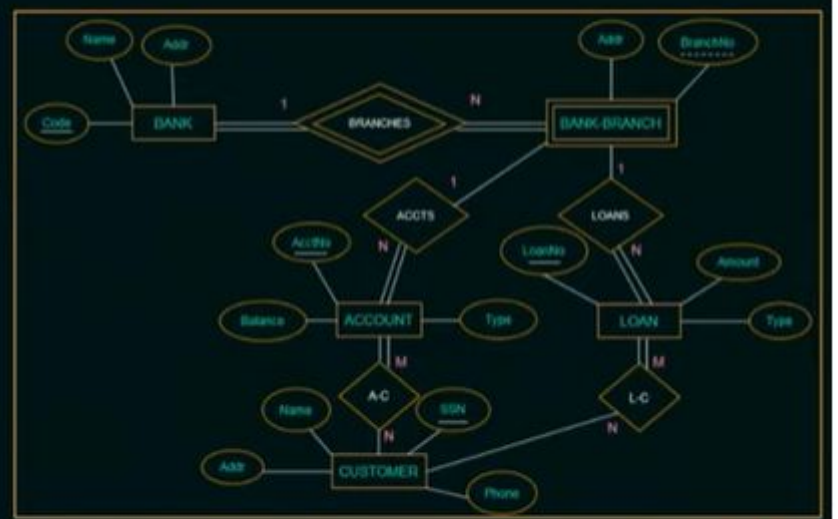
1. Consider the ER diagram shown in the figure for part of a BANK database. Each bank can have multiple branches, and each branch can have multiple accounts and loans.



a. List the (nonweak) entity types in the ER diagram.

❖ Exercises

1. Consider the ER diagram shown in the figure for part of a BANK database. Each bank can have multiple branches, and each branch can have multiple accounts and loans.

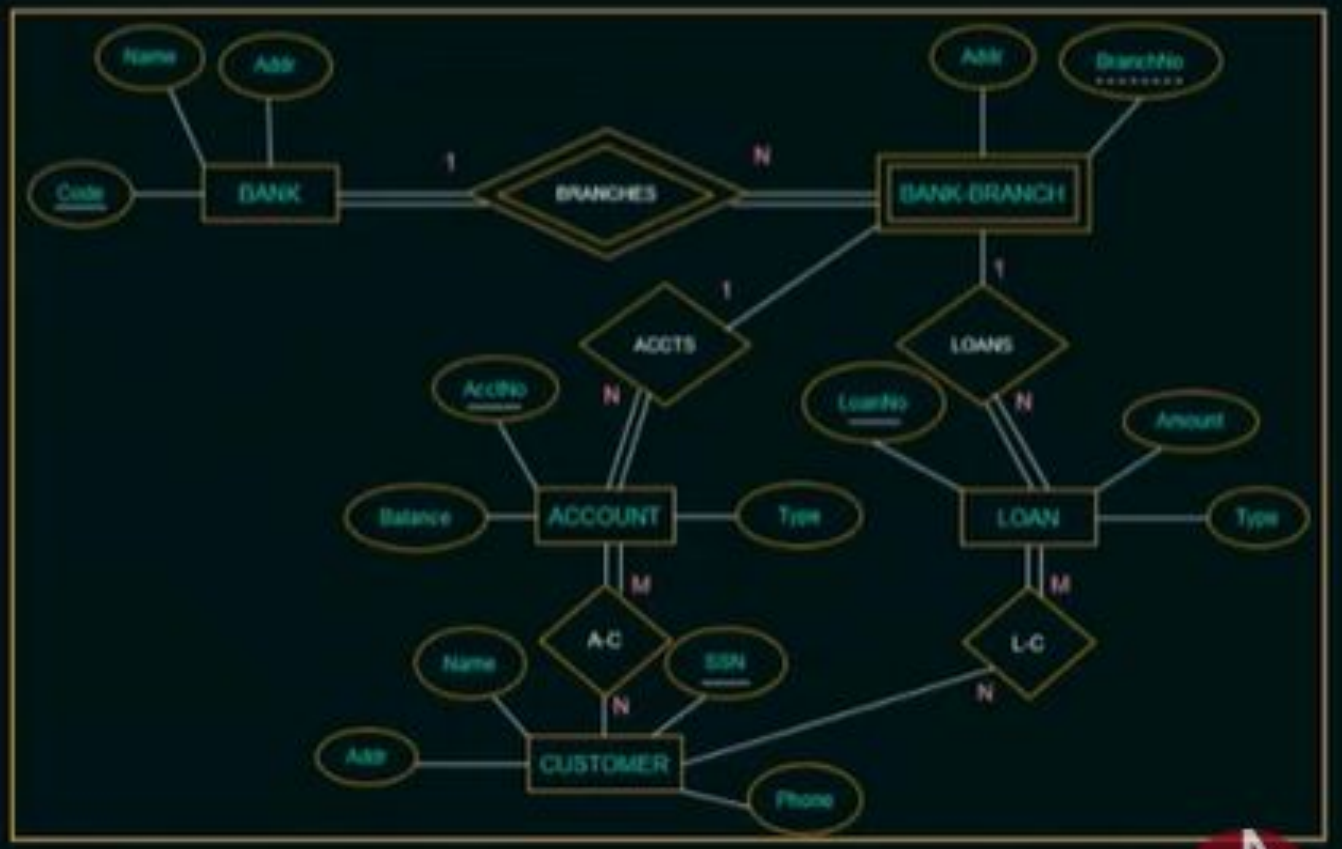


a. List the (nonweak) entity types in the ER diagram.

Sol: Bank, Account, Loan, Customer.

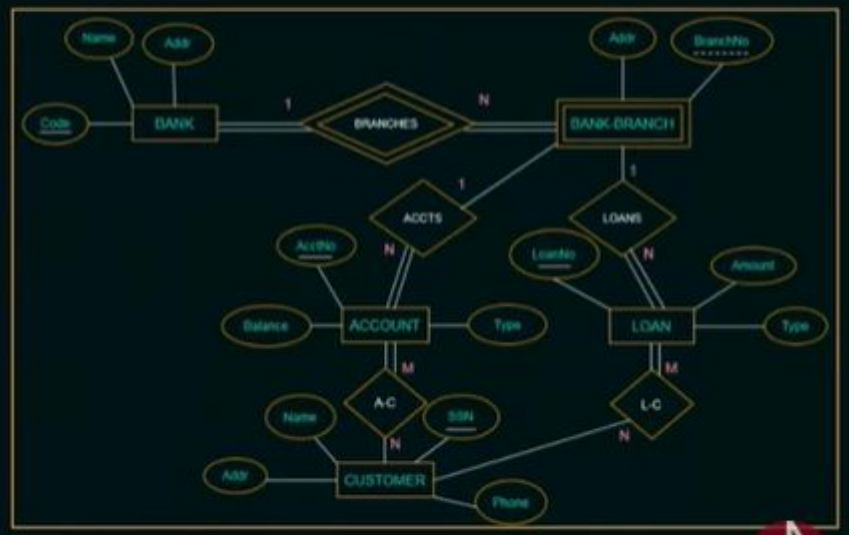
Exercises

b. Is there a weak entity type? If so, give its name, partial key, and identifying relationship.



❖ Exercises

b. Is there a weak entity type? If so, give its name, partial key, and identifying relationship.



Sol: Yes.

Weak Entity Type: Bank Branch,

Partial key: BranchNo,

Identifying relationship: Branches.

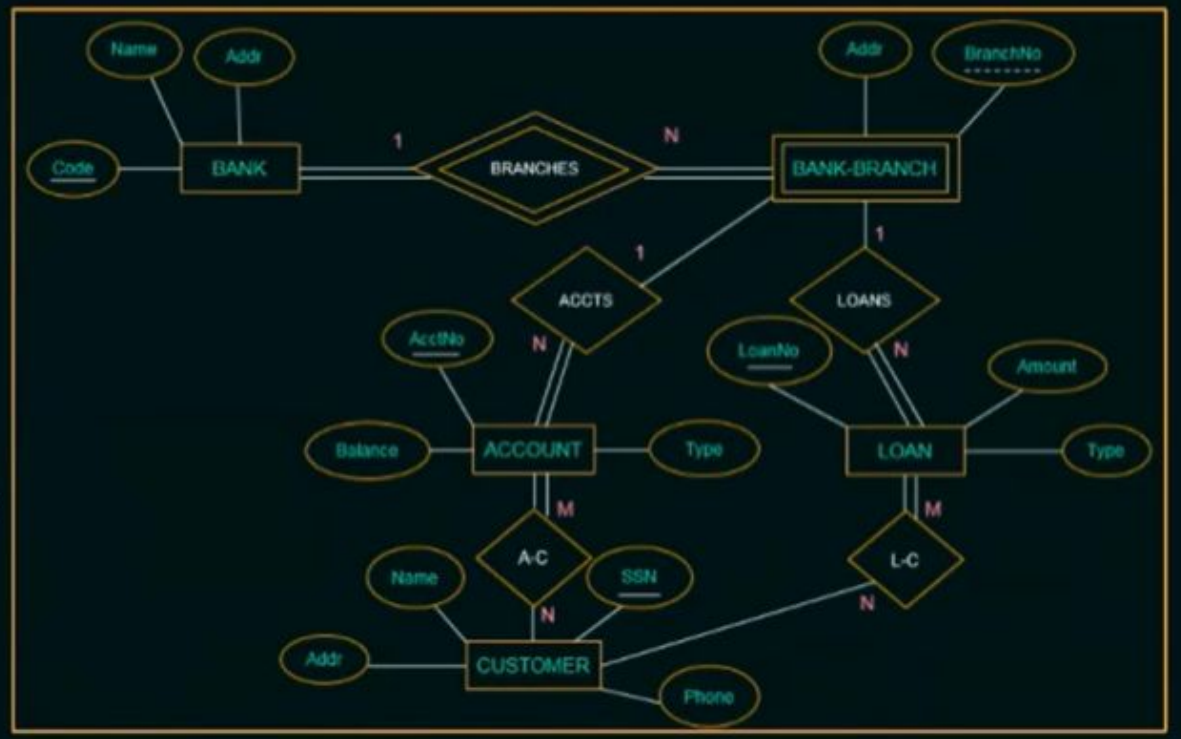
Exercises

c. What constraints do the partial key and the identifying relationship of the weak entity type specify in this diagram?



Exercises

c. What constraints do the partial key and the identifying relationship of the weak entity type specify in this diagram?



Sol: The constraint of the partial key BranchNo is that we need to combine BranchNo with Code (owner entity type key), to uniquely identify Bank-Branch.

Constraint of the identifying relationship:

(i) The weak entity set must have total participation in the identifying relationship set, Branches.

(ii) The identifying relationship between Bank & Bank-Branch must be one to many.

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Exercises

d. List the names of all relationship types, and specify the (min, max) constraint on each participation of an entity type in a relationship type. Justify your choices.



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❖ Exercises

d. List the names of all relationship types, and specify the (min, max) constraint on each participation of an entity type in a relationship type. Justify your choices.

Sol:





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❖ Exercises

e. List concisely the user requirements that led to this ER schema design.



❖ Exercises

e. List concisely the user requirements that led to this ER schema design.

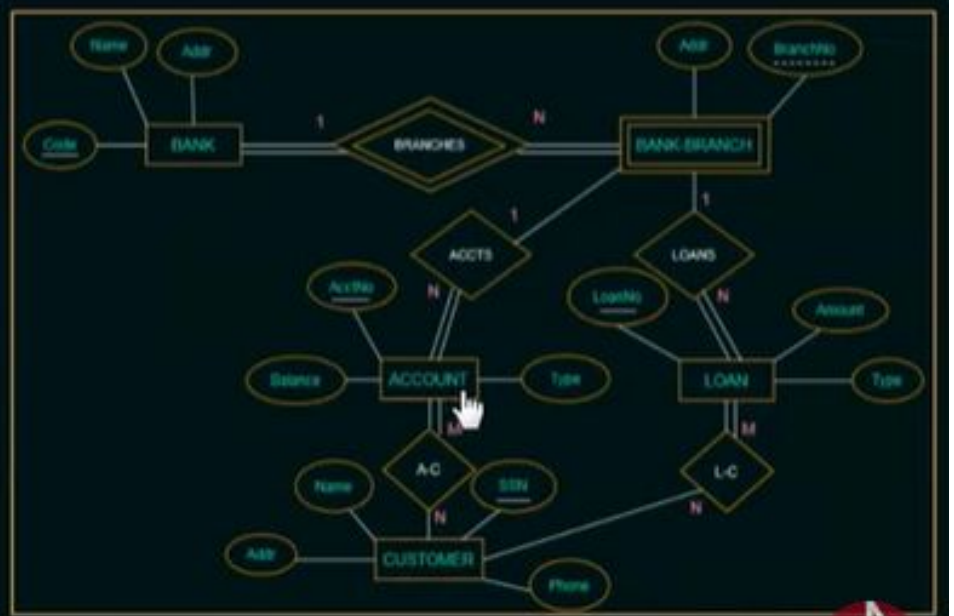
Sol: Requirements can be stated as follows:

(i) Each Bank has a unique Code, Name and Address.

(ii) Each Bank has one or more Branches, each of which has a Branch number & address.

(iii) Each Bank-Branch has zero or more Loans and zero or more accounts.

(iv) Each account has a unique Account Number, Type & Balance. It is related to exactly one Bank-Branch and to at least one Customer.



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❖ Exercises

e. List concisely the user requirements that led to this ER schema design.

Sol: Requirements can be stated as follows:

(v) Each Loan has a unique Loan Number, amount and type. It is also related to exactly one Bank-Branch and to at least one Customer.

(vi) Each Customer has a unique SSN, Name, Address and Phone and is related to zero or more accounts & zero or more loans.



Exercises

1. Consider the ER diagram in the figure. Assume that an employee may work in upto two departments or may not be assigned to any department. Assume that each department must have one and may have up to three phone numbers.

